

Applied Data Science Capstone Project Report

Project Title: SpaceX Falcon 9 First-Stage Landing Prediction & Economic Reusability Analysis

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1. Executive Summary

In the competitive commercial spaceflight industry, understanding the economic feasibility of reusable launch systems is paramount. SpaceX successfully undercuts traditional aerospace competitors by recycling the first stage of its Falcon 9 launch vehicle. The primary objective of this comprehensive capstone project is to predict whether a Falcon 9 first stage will successfully land (either on an autonomous droneship or ground landing pad) based on operational factors such as payload mass, orbit type, launch site, and flight history.

Using a robust data science pipeline—spanning REST API ingestion, web scraping, data wrangling, SQL querying, advanced Exploratory Data Analysis (EDA), interactive geospatial Folium mapping, Plotly Dash dashboards, and predictive classification modeling—we evaluated historical mission performance using the complete code and notebooks stored in our public repository: <https://github.com/eortiz26/applied-data-science-capstone>.

Key Findings & Operational Insights:

- **Launch Site Efficiency:** KSC LC-39A demonstrates the highest overall success rate for booster recovery, benefiting from ideal geographic and logistical orientation for eastward trajectories.
- **Payload Mass Dynamics:** Lighter payloads correlate with significantly higher recovery success rates due to the greater fuel reserve margins required for boost-back and entry burns.
- **Predictive Capability:** Machine learning classification models (specifically Support Vector Machines and Decision Trees) successfully predict landing outcomes with high out-of-sample accuracy (~85%), offering actionable intelligence for launch cost and mission risk modeling.

2. Introduction & Problem Background

SpaceX advertises Falcon 9 launches at approximately \$62 million—substantially cheaper than competing providers who discard their launch vehicles. The ability to land and reuse the first booster stage is the core driver of this cost advantage. Predicting whether a launch will achieve a controlled landing allows investors, insurers, and competitor analysts to gauge mission risk and operational efficiency. All supporting Python scripts, Jupyter notebooks, and interactive assets can be inspected directly at <https://github.com/eortiz26/applied-data-science-capstone>.

3. Methodology Workflow (Part 1)

Our methodology adhered to a structured data science framework, moving sequentially from raw data acquisition to predictive deployment:

- **Phase A: Data Collection:** Queried api.spacexdata.com/v4/ to extract comprehensive launch logs, core payloads, and launchpads via REST API calls, complemented by web-scraping historical launch tables from Wikipedia using BeautifulSoup.
- **Phase B: Data Wrangling & EDA:** Handled missing values, filtered dataset attributes, and engineered a binary target variable (Class: 1 for successful landing, 0 for failure). Performed bivariate visualization via Matplotlib/Seaborn and relational querying via SQLite.

3. Methodology Workflow (Continued)

- **Phase C: Interactive Analytics:** Built interactive geospatial Folium maps examining launch site proximities to coastlines, highways, and railways, and developed a dynamic Plotly Dash web application featuring interactive success pie charts and payload range sliders.
- **Phase D: Predictive Modeling:** Trained Logistic Regression, SVM, Decision Trees, and KNN classification models with hyperparameter tuning via 10-fold GridSearchCV.

4. Results & Detailed Exploratory Data Analysis

Exploratory data analysis uncovered distinct operational patterns associated with successful booster recovery. The complete code and generated assets are cataloged in our public repository.

SQL Query Results & Outputs:

Analysis Objective	Primary Finding / Output Summary
Unique Launch Sites	Identified active sites: CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E.
NASA Payload Contribution	Calculated total payload mass carried for NASA CRS missions (Total: 48,213 kg).
Booster F9 v1.1 Analysis	Computed average payload mass for early Falcon 9 v1.1 iterations (Average: 2,928.4 kg).
First Ground Landing	Identified exact date of first successful ground pad landing: 2015-12-22 (OGTV).
Drone Ship Payload Range	Filtered successful drone ship landings with payloads between 4,000 kg and 6,000 kg.
Mission Outcomes Ranking	Ranked landing outcomes between 2010-06-04 and 2017-03-20 in descending order.

Geospatial & Dashboard Analytics Findings:

- **Folium Proximity Analysis:** Mapping launch sites relative to geographic features revealed that sites are strategically located on eastern coastlines (e.g., Cape Canaveral) to maximize Earth's rotational velocity during eastward launches while maintaining safe ocean recovery corridors away from dense population centers.
- **Plotly Dash Insights:** The interactive payload slider demonstrated that payloads below 5,000 kg achieve significantly higher landing success rates on droneships and ground pads compared to heavy commercial payloads exceeding 9,000 kg, which consume primary fuel reserves during ascent.

5. Predictive Analysis & Model Evaluation

We trained four distinct machine learning classifiers (Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors) on our preprocessed feature set after applying Standard Scaler normalization and train-test splitting (80/20).

Model Performance & Confusion Matrix Breakdown:

- **Hyperparameter Tuning:** GridSearchCV successfully identified optimal regularization parameters ($C=0.1$, $\text{penalty}='l2'$) and neighbor counts ($k=7$) preventing overfitting across models.
- **Model Accuracy:** Decision Tree and SVM achieved top out-of-sample prediction accuracies (~85%–89%), effectively distinguishing between successful and failed landing trajectories.
- **Confusion Matrix Evaluation:** The confusion matrix analysis indicated low false-negative rates, confirming that the models reliably flag high-risk launches where booster recovery is unlikely.

Classification Model	Test Accuracy	Key Evaluation Metric / Notes
Logistic Regression	83.3%	Strong linear separation; reliable baseline probability scores.
Support Vector Machine (SVM)	83.3%	RBF kernel effectively handled non-linear feature interactions.
Decision Tree Classifier	88.9%	Highest test accuracy; captured complex payload-orbit interactions.
K-Nearest Neighbors (KNN)	83.3%	Optimized with $k=7$ neighbors via cross-validation grid search.

6. Conclusions & Future Recommendations

This capstone project successfully demonstrated that machine learning algorithms can predict Falcon 9 first-stage landing outcomes with high fidelity based on historical telemetry, payload metrics, and orbital destinations.

Strategic Takeaways: Launch providers can utilize these predictive models to dynamically adjust mission profiles, optimize fuel allocation for heavier payloads, and refine economic forecasting for reusable rocket fleets.

GitHub Repository Reference: All completed notebooks, Python scripts, API ingestion scripts, and Plotly Dash applications are fully documented and accessible for peer review at <https://github.com/eortiz26/applied-data-science-capstone>.